Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/22/2021
PI Name:	Pecaut, Michael Ph.D.		
Project Title:	Impact of Spaceflight on Primary and Secondary Antibody Responses		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	mpecaut@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8372
Organization Name:	Loma Linda University		
PI Address 1:	Division of Biomedical Engineering Sciences, Department of Basic Sciences		
PI Address 2:	11175 Campus St, Chan Shun Pavilion, A-1016		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	07/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Loftus, David	**Contact Phone:**	650-604-1011
Contact Email:	david.j.loftus@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 7/31/2022 per NSSC information (Ed., 10/14/21) NOTE: End date changed to 7/31/2021 per F. Hernandez/ARC (Ed., 6/4/21) NOTE: End date changed to 3/30/2021 per NSSC information (Ed., 9/10/20) NOTE: End date changed to 09/30/2020 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 11/30/2018 per NSSC (Ed., 10/10/17) NOTE: End date changed to 11/30/2017 per NSSC (Ed., 6/3/16)		
Key Personnel Changes/Previous PI:	June 2019 report: Co-Investigator Stephen K. Chapes, Ph.D., retired from his position at Kansas State University (KSU) and is now serving as a CoI consultant on this grant. September 2018 report: Fomer CoI Daila Gridley, Ph.D., is no longer participating in the project.		
COI Name (Institution):	Chapes, Stephen Ph.D. (Retired (formerly from Kansas State University))		
Grant/Contract No.:	NNX13AN34G		
Performance Goal No.:			

Performance Goal Text:

The spaceflight environment influences immune cell distribution, ex vivo responses to antigen, and cytokine expression. However, few studies have characterized the impact of the spaceflight environment on the host immune response following an in vivo immune challenge, particularly those involving B cells and antibody production. The International Space Station (ISS) provides an ideal laboratory within which investigators can explore this unique environment and its impact on the host response. Through the use of a mouse model (with an immune system that closely parallels that of humans), we will establish a link between exposure to the spaceflight environment and decrements in immune system activation. This study is significant because it will directly test the hypothesis that host immunological health will be compromised in space.

Task Description:

 On-orbit Flight Activities, Procedures, and Timeline

Mice will receive health checks upon unloading and transfer from the Transporter to the Habitat after capsule docking no later than 24 hrs after hatch opening. Animal health will then be monitored daily via continuous video recording from one camera at a time from all of the Habitats on-orbit (including other investigations which may be running concurrently) and down-linked to Ames science staff. Ground personnel shall have the option of programming or manually controlling which camera is being down-linked and recorded at any given time. Water levels will be checked every 3-4 days. After a minimum of 14 days after animal transfer from the Transporter to the Habitat, all 20 mice will be moved to the MSG and receive intraperitoneal (i.p.) injection of vehicle (5 mice), Tetanus Toxoid (TT) vaccine (5 mice), CpG ODN adjuvant (5 mice), or TT vaccine + CpG ODN adjuvant (5 mice). The dose of TT and/or CpG ODN will be 10 microg/ mouse and 40 microg/ mouse, respectively. Pre-mixed doses will be prepared preflight in 100 microl saline and stored at -20 to -30°C until the appropriate time point. All injections and blood draws will use a 1 mL 26G 3/8” tuberculin syringe. All injections will be i.p.

Before use, the CpG adjuvant must be stored at -20°C or colder until use and the TT vaccine must be stored at 4°C. At 14 +/- 2 days after injection, mice will be transferred to the MSG and sacrificed via an i.p. injection of 100/10 mg/kg ketamine/xylazine mixture, followed by exsanguination by cardiac puncture. Immediately following the blood collection, the spleens from the mice will be removed, sectioned into halves, and placed in a frozen environment (minimum temperature -80°C) until returned to Earth. Following spleen collection, the remains of the mouse carcass will be wrapped in tin foil, put in a zip-lock bags and placed in a -80°C environment (or colder) to preserve the tissues for future biospecimen sharing. Blood must not be allowed to sit at ambient temperature for longer than 1 hour post-collection; if operational requirements require the blood collected to be stored in excess of 1 hour before centrifugation, the blood must be transferred to a 4°C environment until centrifugation, for a period not to exceed 4 hours. Blood collected will be separated into solid and serum/plasma fractions by centrifugation and frozen at -80°C until returned to Earth. Storage times for both spleen and blood of up to 6 months at -80°C are acceptable.
 <table border=1 cellspacing=0 cellpadding=0 align=center width=672> <tr > <td width=91> Time </td> <td width=252> Activity & Estimated Time Required </td> <td width=76 > Thermal Req. </td> <td width=253 > Comments, Location & Special Considerations </td> </tr> <tr > <td width=91 > Launch to L + 45 days </td> <td width=252 > Animal health checks via video monitoring; water level checks every 3-4 days </td> <td width=76 > 20 – 30°C </td> <td width=253 > Video downlink required for health checks </td> </tr> <tr> <td width=91 > Transfer to Habitat + 14 d </td> <td width=252 > All 20 mice will be transferred to habitat </td> <td width=76 > 20 – 30°C </td> <td width=253 > Requires transfer to and from Habitat with MTB; </td> </tr> <tr > <td width=91 > Injection + 14d +/- 2 d </td> <td width=252 > At specific time points, all 20 mice will be removed from AEM and given appropriate injection </td> <td width=76 > 20 – 30°C / -80°C </td> <td width=253 > Required transfer to MSG for injections </td> </tr> <tr > <td width=91 > Dissection + 0 d </td> <td width=252 > Mice euthanized, blood drawn and centrifuged and spleen sectioned and frozen. Blood and spleen samples kept at -80^oC until delivered to laboratory of PI </td> <td width=76 > -80°C </td> <td width=253 > Required transfer to MSG for dissection, processing of blood in the centrifuge, and storage of samples at -80°C for up to 6 months. </td> </tr> </table>

  Hardware Housekeeping Operations: Crew-performed

Daily hardware checks will occur. Habitats will be kept at a regular 12hr:12hr day:night light cycle to match the day/night programmed to be synchronous with the crew’s day/night cycle upon arrival. Water refill of habitats shall be performed. To prevent rodent food bars from becoming too hard or soiled, they will be replaced with a fresh food bar plate (from 4°C storage) at a time to not exceed 30 days after the introduction of mice to a Habitat.

Ground Activities Accompanying On-orbit Activities and Timeline

Ground controls will be run following spaceflight environmental parameters in orbital environmental simulator(s) (OES) chambers at KSC ACF. Delay between launch and start of environmental controls may be up to 8 days, depending on the logistics and capability of the telemetry system. Mice will be housed in flight Habitats with RFB diet and lixit water to match on-orbit groups. The operations done on ground will be conducted to match the same relative time points during the day/night cycle of the animals on-orbit, so that procedures on ground control mice are conducted at the same points in the animals’ circadian rhythm.

Each experimental group for ground controls will be doubled to improve the statistical significance of the experiment with a total of 40 mice (4 groups of 10). Therefore, 2 Transporters will be required for the flight in orbital environmental simulator(s) (OES) that have been programmed with telemetry data from the flight. At the same relative time point (accounting for delay in starting ground controls after flight) the mice are transferred from the Transporter to the Habitat, the 40 ground control mice will be transferred from the 2 Transporters into 4 Habitats, and maintained in OES programmed with telemetry data from the ISS. OES shall be programmed to mimic the following conditions experienced by the Transport and then the Habitat: CO2, O2, temperature, and humidity.

Injection and dissection must be performed as close as possible to how they are done on-orbit. In particular, the time of

day that mice are sacrificed and dissected on the ground should match the same point in the circadian rhythms of the mice on-orbit. However, dissections or ground control mice can be conducted on standard laboratory benches/tables with chairs that are more ergonomic to the operators. For ground controls, dissection pins or needles may be used to pin down mice for dissection, as can spray bottles of wetting the mice before dissection. Flight hardware is required when it may affect the tissue dissection or preservation, such as having flight cold blocks in mini cold bags for tissue freezing.

Mice will receive health checks daily via video in the Habitat and water levels will be checked every 3-4 days. After a minimum of 14 days in the OES, all 40 mice will receive an i.p. injection of vehicle (10 mice), TT vaccine (10 mice), CpG adjuvant (10 mice), or TT vaccine + CpG adjuvant (10 mice). At 14 +/- 2 days after injection, mice will be sacrificed via an i.p. injection of 100/10 mg/kg ketamine/xylazine mixture, followed by exsanguination by cardiac puncture. Immediately following the blood collection, the spleens from the mice will be removed, sectioned into halves, and placed in a frozen environment (minimum temperature -80°C) until returned to Earth. Duration for which blood is stored at ambient or 4°C shall be matched as closely as possible to the timelines the same procedures will be performed on-orbit. Blood collected will be separated into solid and serum/plasma fractions by centrifugation and frozen at -80°C until returned to the laboratory of the PI. Storage times for both spleen and blood of up to 6 months at -80°C are acceptable.

<table class=MsoNormalTable border=1 cellspacing=0 cellpadding=0 align=center width=672 > <tr > <td width=145 > Time </td> <td width=186 > Activity </td> <td width=66 > Thermal Req. </td> <td width=275 > Comments, Justification, Special Considerations </td> </tr> <tr > <td width=145 > Beginning of ground controls to +45 days </td> <td width=186 > Animal health checks; water level checks every 3-4 days </td> <td width=66 > 20 – 30^oC </td> <td width=275 > Video downlink required for health checks </td> </tr> <tr > <td width=145 > TBD, depending on delay in ground controls and on-orbit timelines </td> <td width=186 > At specific time points, all 40 mice will be removed from AEM and given appropriate injection </td> <td width=66 > 20 – 30^oC </td> <td width=275 > Transfer from cages to a regular lab bench and back w/ standard cages. </td> </tr> <tr > <td width=145 > TBD, depending on delay in ground controls and on-orbit timelines </td> <td width=186 > Mice euthanized, blood drawn and centrifuged and spleen sectioned and frozen </td> <td width=66 > -80^oC </td> <td width=275 > Processing of blood in a centrifuge, and storage of samples at -80^oC </td> </tr> </table>

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

According to the National Institute of Aging (NIA), “Life expectancy nearly doubled during the 20th century with a ten-fold increase in the number of Americans age 65 or older.” Furthermore, “there are approximately 35 million Americans age 65 or older, and this number is expected to double in the next 25 years.” Consequently, there is a rapidly growing need to understand the mechanisms underlying age-associated illness. Indeed, in the last decade, research has repeatedly demonstrated an impact of aging on immune function. Low-grade increases in circulating inflammatory cytokine secretion are often associated with senescence. Furthermore, studies suggest that these disruptions are likely due to decrements in cellular repair mechanisms and/or increases in antigenic load. Given this disruption in immune function, it is perhaps not surprising that many of the diseases associated with aging — such as type 2 diabetes, rheumatoid arthritis, Parkinson’s disease, Alzheimer’s disease, cardiovascular disease — have strong inflammatory underpinnings.
There are strong indications that the physiological effects of spaceflight are also similar to those seen in senescence. For instance, evidence suggests that age-associated enhancements in inflammatory cytokines are linked to changes in adipose tissue and energy balance. Similar links between metabolism, adipose tissue, and inflammation have also been shown to manifest in astronauts and animal models after exposure to spaceflight. We propose to use long-term exposure to the spaceflight environment to model the effects of aging on immune function. Although these initial studies involve mice, the next obvious step in this work will be to work directly with astronauts. Because astronauts tend to be extremely healthy, these effects will not be influenced by confounding factors such as clinically relevant, age-related disease. Also, because the effects of spaceflight on immunity are temporary, generally returning to baseline levels post-flight, this will ultimately give us some insight into whether or not the effects of aging are, indeed, reversible.

Although many different aspects of immune function are influenced by both aging and spaceflight, these experiments will focus on B cells and the generation of antibodies. Specifically, we will use next generation RNA sequencing techniques to characterize the B cell Ig heavy chain in animals that are challenged with tetanus toxoid. The Ig heavy chain is the component of antibodies responsible for identifying specific sections of the tetanus toxoid, thereby coordinating subsequent immune responses. Comparisons will be made between the B cell repertoires generated on ISS and on Earth. The changes in B cell Ig heavy chain usage will provide indications whether the spaceflight environment affects the B cells that respond to antigen. This, in turn, will help us understand the impact of aging on B cell function.

We anticipate that we will obtain several data sets which have relevance to spaceflight and/or have general significance to understanding host immune responsiveness. These data include: 1) Data that show that spaceflight either maintains or alters the basic distribution of B cells based on Igh transcripts; 2) Information that demonstrates how immunization with an immunogen, commonly used in human medicine, alters the basic distribution of B cells based on Igh transcripts in the absence of the spaceflight variable; 3) Determinations how activation of B cells through Tlr9 receptors alters the basic distribution of B cells based on Igh transcripts.

We will ascertain if activation of B cells through Tlr9 receptors can reverse any impact that may occur during spaceflight.

Task Progress:	COVID-19 Related Delays: Loma Linda University (LLU) went on COVID-related lockdown in March 2020. At that time, all staff were strongly encouraged to work from home if possible. Much of the work associated with this grant involved tissue processing and data analysis. While the data analysis could continue (we were able to get permission for our post doc to connect remotely to our server), the tissue processing was put on hold. This coincided with a shutdown at the Kansas State University Integrated Genomics Facility for both COVID-related issues and a temporary university-wide power failure that halted processing for several weeks. Due to the uncertainty related to COVID, both of our post docs (Drs. Trisha Rettig and John Tan) opted to move on to other positions. Dr. Rettig left for a position in industry on 10/5/2020. Dr. Tan moved on to a faculty position within LLU on 6/30/2020. Similarly, our research technician, Nina Nishiyama, moved on to go to graduate school at the University of North Carolina on July 17, 2020. Currently, all the tissue processing at LLU is now complete and all our samples have been shipped to the Kansas State University (KSU) Integrated Genomics Facility. To date, we have run 14 of at least 40 samples. Once these samples are processed, all that remains is the completion of data processing/analysis and manuscript preparation. While Dr. Rettig has offered to continue working on this project from her new position, it is essential to bring on a new graduate student or post doc to complete this work. We recently received an COVID-19 augmentation from Space Biology complete the work.
Bibliography Type:	Description: (Last Updated: 08/04/2025)
Articles in Peer-reviewed Journals	Rettig TA, Tan JC, Nishiyama NC, Chapes SK, Pecaut MJ. "An analysis of the effects of spaceflight and vaccination on antibody repertoire diversity." Immunohorizons. 2021 Aug 25;5(8):675-86. https://doi.org/10.4049/immunohorizons.2100056 ; PMID: 34433623 , Aug-2021
Articles in Peer-reviewed Journals	Oyefeso FA, Muotri AR, Wilson CG, Pecaut MJ. "Brain organoids: A promising model to assess oxidative stress-induced central nervous system damage." Dev Neurobiol. 2021 Jul-Aug;81(5):653-70. https://doi.org/10.1002/dneu.22828 ; PMID: 33942547; PMCID: PMC8364474 , Jul-2021